

Derivatives Overlay Program for a Rs 78 Cr Concentrated Indian Equity PMS Mandate (v2, Live Data)

A CFA Level III Applied Project

Designed, priced, and risk-managed a full derivatives overlay — protective puts/collars, covered calls, an interest-rate swap duration overlay, and a currency risk-reversal structure for USD-denominated LRS holdings — for a simulated SEBI-registered boutique PMS with concentrated, promoter-heavy holdings (Rs 78 Cr AUM, 14 HNI clients, single-stock concentration up to 11%).

v2 update: replaced simulated equity position data with live QuantsApp Pro+ data (spot, IV, IV percentile) for all 10 sleeve names, and discovered — by directly checking rather than assuming — that the four originally-planned protection-sleeve archetypes were not NSE F&O-eligible at all. Screened and substituted four real, currently-optionable, promoter-driven names (SRF, Bharat Forge, Kalyan Jewellers, Polycab) against a documented four-point criteria set (F&O eligibility, concentration weight, promoter/event-sensitivity, relative IV), then recomputed the entire model against their live market data.

Built a Black-Scholes-Merton pricing engine from first principles (no third-party options library), an objective-to-strategy decision matrix, and an implied-volatility-percentile hedge-timing rule that now reads QuantsApp's real IV Percentile directly rather than a model-only approximation. Quantified the program end-to-end: **27.5 bps/yr gross hedging cost against a 40–60 bps Investment Committee budget, funded more than fully by 177.3 bps/yr of covered-call income (richer than the original placeholder, because real current IVs — e.g. Infosys ~34% — price higher) — a net cash-flow-positive program** — cutting modeled stress-scenario drawdown by 177–888 bps depending on severity.

Stack: Python (NumPy, SciPy `optimize`, pandas, Matplotlib), Excel (live formulas, not hardcoded values), Jupyter, QuantsApp Pro+ (live equity/options market data).

Deliverables: report with IC memo, risk framework, and a documented stock-selection methodology · executed Jupyter notebook · Excel workbook with live Black-Scholes formulas · 7-chart visual package · QuantsApp Pro+ source screenshots · GitHub-ready repository.